Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models

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B.Tech CSE Semester 8 Final Evaluation

Agenda

1. Methodology
2. Architecture
3. Implementation
4. Results
5. Future Work
6. Conclusion

Methodology

Dataset Creation and Augmentation

- **Data Source:** Government Medical College (GMC), Kozhikode
- **Raw Data:** 6,500 **patient reports** (semi-structured PDFs + endoscopic images)
  - Each report includes **4 endoscopic images**
- **Data Augmentation:**
  - Applied **rotation and random translation** to every image
  - Purpose: increase dataset size and improve model robustness
  - **Approximate scale:**
    - 6,500 reports $\times$ 4 images $=$ 26,000 base images
    - Rotation alone multiplies by 4 $\rightarrow$ over 100,000 images
    - Translation variations further expand the pool
- **Text Data:**
  - Final Q&A pairs derived from report content: **32,545 unique samples**
  - Data split: 30,917 training / 1,627 validation
- **Multimodal Association:** Each Q&A pair linked to original + augmented endoscopic images Chen and et al. [2024], Zhang and et al. [2024]

Figure 1: Custom Dataset Workflow

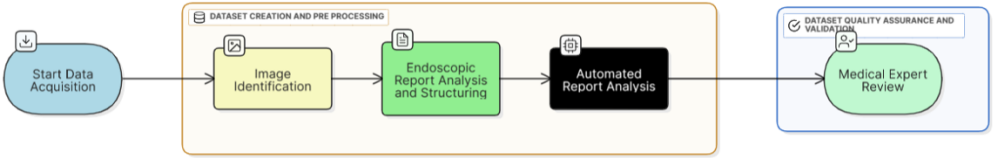


Figure: Data Workflow: From 6,500 patient reports to 32,545 Q&A pairs Chen and et al. [2024], Zhang and et al. [2024]

Data Augmentation: Rotation

- Each original endoscopic image is rotated by **0°, 90°, 180°, and 270°**
- Produces exactly **4 variants** per image
- Clinically safe: tissue textures and lesion appearances remain unchanged
- **Effect:** $4\times$ structured expansion of the base image set

Data Augmentation: Stochastic Translation

- **Sampling:** Bernoulli process with $p = 0.2$ selects approx. 20% of rotated images for translation
- **Shift model:** Translation vector (τ_x, τ_y) drawn from a truncated Gaussian:

$$\tau_x \sim \mathcal{N}(0, (0.05\,W)^2), \text{ clamped to } [-0.15\,W, +0.15\,W]$$
$$\tau_y \sim \mathcal{N}(0, (0.05\,H)^2), \text{ clamped to } [-0.15\,H, +0.15\,H]$$

- **Overlap guarantee:**

$$\text{Overlap Fraction} = \left(1 - \frac{|\tau_x|}{W}\right) \times \left(1 - \frac{|\tau_y|}{H}\right) \geq (1 - 0.15)^2 = 72.25\%$$

- **Result:** Controlled positional variation mimicking endoscopic camera jitter, without harming diagnostic content.

Data Augmentation Examples

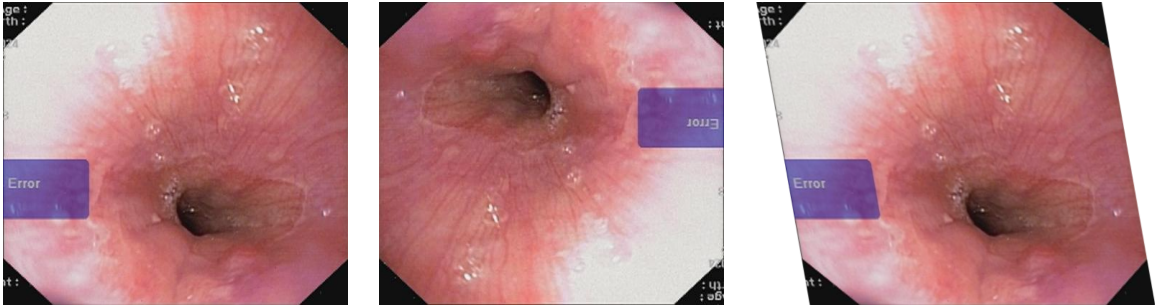
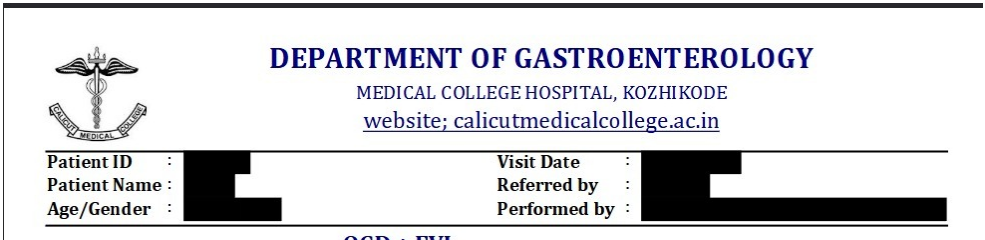


Figure: Left: Original Middle: Rotated (180°) Right: Randomly translated

Patient Data De-identification & Ethical Compliance

- **Regulatory Framework & Automated Script:**
  - Strict adherence to the **Digital Personal Data Protection (DPDP) Act**, supported by a formal MoU between GMC Kozhikode and NIT Calicut
  - **Redaction approach mirrors rigorous de-identification used in recent high-profile document releases (e.g., Epstein files)** – all personally identifiable fields irreversibly blacked out to ensure zero traceability
  - **Header Field Redaction:** Automatic blackout of Patient ID, Name, Age/Gender, Visit Date, UGI/OGD Numbers
- **Result:** Irreversible anonymization of all sensitive patient information before data enters the pipeline



DEPARTMENT OF GASTROENTEROLOGY

MEDICAL COLLEGE HOSPITAL, KOZHIKODE

website; calicutmedicalcollege.ac.in

Patient ID :	[redacted]	Visit Date :	[redacted]
Patient Name :	[redacted]	Referred by :	[redacted]
Age/Gender :	[redacted]	Performed by :	[redacted]

Report Analysis and Preprocessing

Medical Entities Extracted: Chen and et al. [2024], Zhang and et al. [2024]

- Pathological findings and characteristics
- Diagnostic conclusions
- Procedural techniques and instruments

Processing Method:

- Natural language processing techniques
- Extraction of critical medical elements
- Foundation for Q&A pair generation

Automated Question-Answer Pair Generation

Multi-Stage Process: Hicks [2024], Pogorelov and et al. [2024]

1. **Medical Entity Extraction** – Key concepts and diagnoses
2. **Question Template Design** – Diagnostic, anatomical, pathological, procedural, comparative
3. **Automated QA Generation** – Multiple question variations per concept Liu et al. [2024], Mallick [2024]
4. **Answer Synthesis** – Accurate answers from reports + medical knowledge

Result: Diverse, clinically relevant question-answer combinations.

Dataflow Pipeline and Methodology

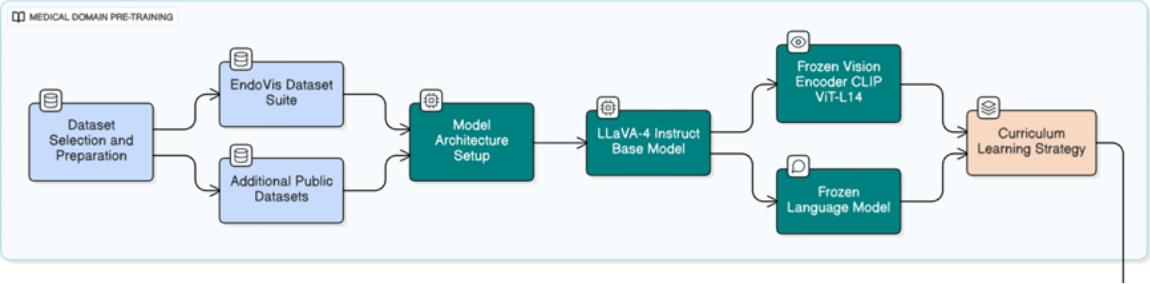


Figure: Stage 1: Pre-training.

Dataflow Pipeline and Methodology

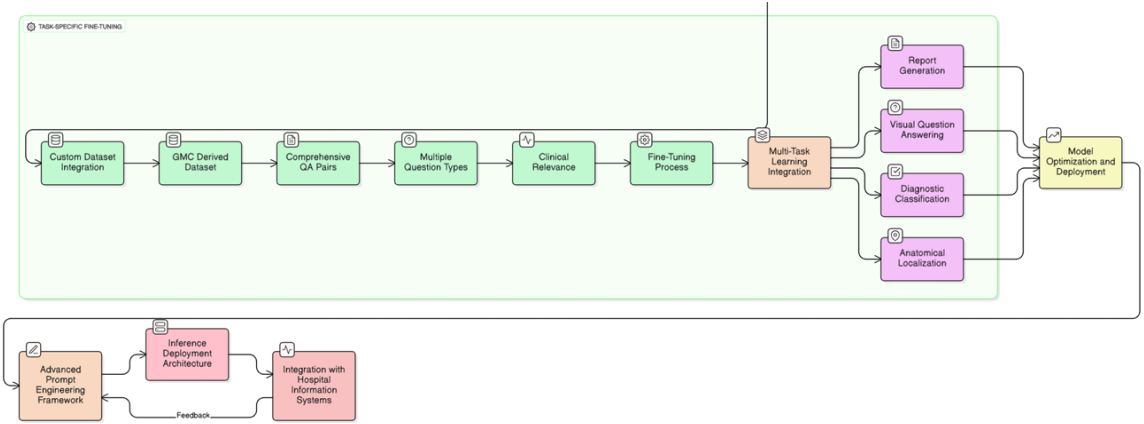


Figure: Stage 2: Fine-tuning

Architecture

Two-Stage Training Framework

Stage 1: Medical Domain Pre-Training Leaman and Lu [2024], Liu et al. [2023], Li et al. [2023]

- **Objective:** Establish medical concept alignment
- **Base Model:** LLaVA-4 with frozen CLIP ViT-L/14
- **Trainable:** Only multimodal projection layer
- **Datasets:** EndoVis 2017/2018 + Endo700k (700,000+ images) Chaitanya et al. [2024], Speidel and et al. [2024], Maier-Hein and et al. [2024]

Stage 2: Task-Specific Fine-Tuning Leaman and Lu [2024], Li et al. [2023]

- **Objective:** Instruction tuning for diagnosis and report generation
- **Dataset:** Custom GMC-derived Q&A pairs (32,545 samples)
- **Strategy:** Parameter-Efficient Fine-Tuning (LoRA) Dutta [2024], UBIAI Team [2024]
- **Unfrozen:** Projection layer + language model parameters

System Architecture

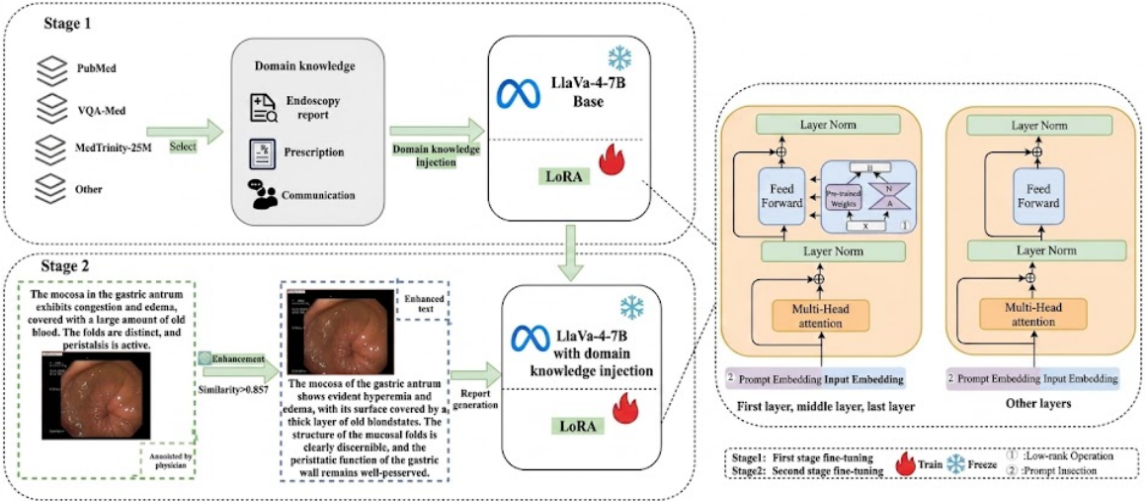


Figure: High-level architecture of the proposed vision-language model for endoscopic report generation.

Model Architecture – LLaVA

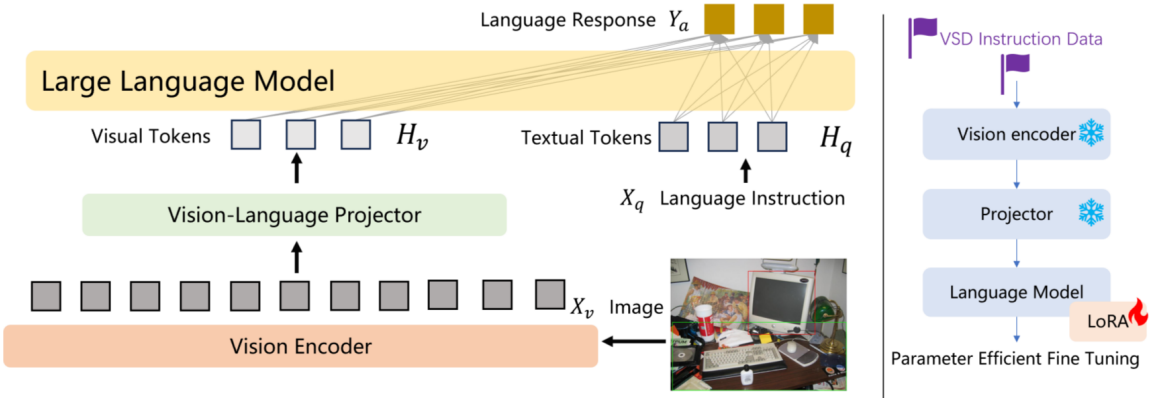


Figure: LLaVA architecture: vision encoder + projection layer + language model for multimodal understanding.

CLIP-ViT Architecture

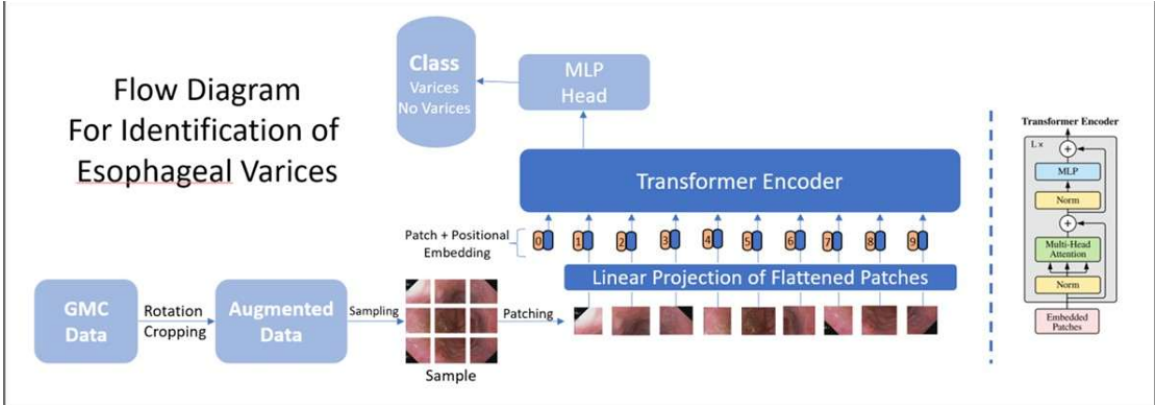
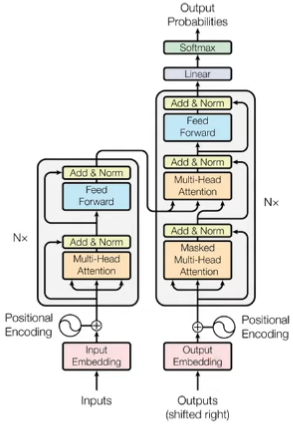
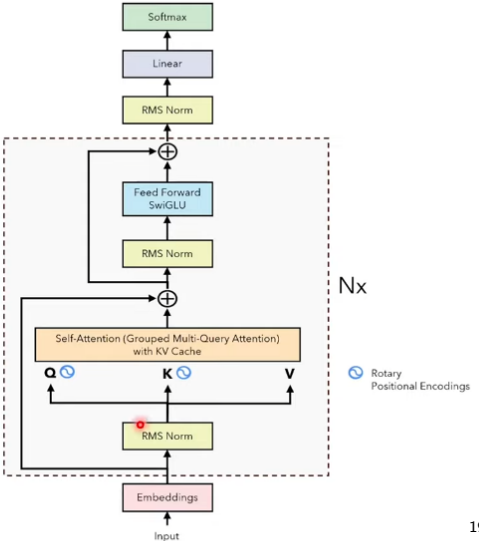


Figure: CLIP-ViT-L/14 vision encoder: patch embedding, transformer layers, and projected visual features.

Transformer vs LLaMA





Implementation

Refining Generation: Implementing System Prompts

Initial Challenge:

- Early model iterations produced verbose, unstructured, and redundant reports that did not match standard clinical formatting.

Implemented Solution (Dataset Reformatting):

- Integrated rigorous **System Prompts** directly into the training dataset.
- **Role Assignment:** Instructed the model to assume a specific clinical persona (e.g., "You are an expert gastroenterologist...").
- **Output Constraints:** Provided explicit instructions to prioritise direct, concise, and structured answers mirroring real GMC Kozhikode reports.

Result: A massive reduction in unnecessary text, yielding highly concise and structured outputs; further refinement improved diagnostic accuracy.

Qualitative Impact: Verbose vs. Concise Reports

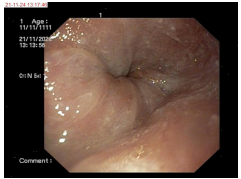
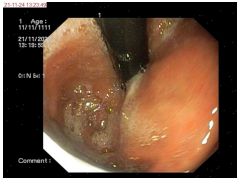
Previous Generation
(Verbose & Redundant)

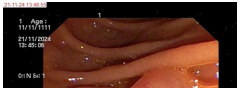
Medical Endoscopy Report

Patient Information

Patient Name: ABDUL MANAF
Patient ID: ABDUL MANAF_86376_RP241121_015818
Report Date: 2025-12-04 12:55:59

Endoscopy Images



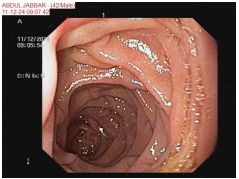
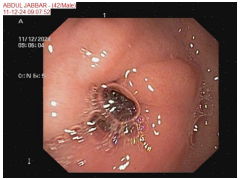
Current Generation
(Concise via System Prompt)

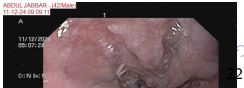
Medical Endoscopy Report

Patient Information

Patient Name: ABDUL JABBAR
Patient ID: ABDUL JABBAR_84105_RP241211_091205
Report Date: 2026-03-06 13:45:39

Endoscopy Images



Qualitative Impact: Verbose vs. Concise Reports

Previous Generation
(Verbose & Redundant)

Scope Passed Upto:	D2 mucosa without ulcers/ erosions/ luminal narrowing .Scope passed upto D3 mucosa without any difficulty .Biopsy taken from D2 and D3 mucosa for HPR and CBNAAT from D2 and D3 mucosa . D1 mucosa was edematous and erythematous with superficial ulcers . D1 was not entered further due to poor patient tolerability . D1 mucosa was edematous and erythemat
Premedication:	10% XYLOCAINE ANTRUM AND PYLORUS WITH SURROUNDING NODULARITY AND D1 NODULARITY FOR BIOPSY TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BARIUM SWALLOW IF FURTHER EPISODE/HB FALL BIOPSY TAKEN FROM NODULARITY AND D1 NODULARITY FOR CBNAAT TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BAR
Esophagus:	Normal mucosa without ulcers/ erosions/ luminal narrowing with normal peristalsis and distensibility noted .No foreign body seen in the esophagus .No esophageal varices seen .No altered blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No altered blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No altered blood seen in the lumen .No luminal blood seen
Ge Junction:	40CMS and edematous and erythematous mucosa noted.Biopsy taken .No nodularity or ulcerations noted. No tear noted at present. No salivary pooling. No difficulty in swallowing. No foreign body seen. No luminal blood seen. No active ooze seen. No hemoptysis. No melena. No hematemesis. No hematuria. No proteinuria. No renal papilla. No ureter. No ampulla. No pyloric opening. No duodenal ulcer. No
Stomach:	No stasis . Normal peristalsis and distensibility noted . No ulcer/ erosions noted . Normal blood in lumen . No active bleed at present . Normal mucosal folds and contours noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal
Fundus:	Normal mucosa without ulcers/ erosions/ luminal narrowing with normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal

Current Generation
(Concise via System Prompt)

Stomach:	
Fundus:	No varices
Body:	Mosaic pattern of mucosa seen
Antrum:	Normal
Pylorus:	Normal
Duodenum:	D1: Normal | D2: Normal
Rut:	Mosaic pattern of mucosa seen
Biopsy:	Not taken
Impression:	GRADE II X3 COLUMNS OF ESOPHAGEAL VARICES WITHOUT RCS SEVERE PHG

Report generated by Medical LLaVA AI on 2026-03-06 at 13:45:39

The “Blackbox” Problem & Our Solution

The Initial Blackbox:

- The model would output a final diagnosis directly from an image without revealing its reasoning, making it hard to trust or debug.

Our Solution – Chain-of-Thought Integration:

- Augmented the training Q&A pairs with **step-by-step reasoning chains**.
- Each chain forces the model to structure its answer as: **anatomical identification** $\rightarrow$ **visual observation** $\rightarrow$ **diagnosis**.
- During training, the model learns to mimic this transparent reasoning process.

Outcome: The model now explicitly explains *what it sees* and *why it concludes something*, turning a black-box predictor into an **interpretable diagnostic assistant**.

Chain-of-Thought (CoT) in Endoscopy

Before CoT Implementation (Baseline Model):

- **Input:** [Image] "What is the diagnosis?"
- **Output:** "Grade II Esophageal Varices." *(correct but no justification)*

After CoT Implementation (Our Model):

- **Input:** [Image] "Analyze the image step-by-step and provide a diagnosis."
- **Output:**
  1. **Observation 1:** I observe the lower third of the oesophagus.
  2. **Observation 2:** There are swollen, tortuous submucosal veins present.
  3. **Observation 3:** The veins occupy less than 1/3 of the lumen, with no red colour signs.
  4. **Conclusion:** Therefore, the diagnosis is Grade II Esophageal Varices.

Impact: Explicit reasoning enhances clinical trust, allows verification, and helps identify subtle mistakes – a crucial feature for real-world deployment.

Fine-Tuning Configuration Details

Vision Encoder: CLIP-ViT-L/14

- Image resolution: 336×336 pixels
- Patch size: 14×14 (24×24 grid)
- Total vision tokens: 577 per image (576 patches + CLS)

Training Hyperparameters:

- **Precision:** bfloat16 (BF16)
- **Optimizer:** AdamW with 8-bit quantization
- **Learning Rate:** $2e^{-4}$ with cosine scheduler, warmup 0.1
- **Batch Size:** 2 micro-batch, 4 accumulation steps = 8 global
- **Attention:** Flash Attention 2
- **Epochs:** 2 epochs, 7,730 steps, 253M tokens

Training Configuration Rationale

- **CLIP-ViT-L/14** – pre-trained on 400M pairs; robust medical visual features; L size balances capacity vs. speed; 14×14 patch grid captures both anatomy and small lesions.
- **336×336 resolution** – standard for LLaVA; sufficient detail without excessive memory.
- **577 vision tokens** – each patch token carries localised information, enabling the language model to "see" global and local structures.
- **bfloat16** – halves memory vs. FP32, no performance loss.
- **AdamW + 8-bit** – stable weight-decoupled convergence; 8-bit reduces optimizer memory 75%.
- **LR $2e^{-4}$, cosine + warmup** – low peak prevents catastrophic forgetting; warmup stabilises early steps.
- **Batch size 8 (accumulated)** – simulates larger batch under GPU memory limits.
- **Flash Attention 2** – memory-efficient exact attention; 5-10× VRAM reduction for long sequences.
- **2 epochs** – sufficient for LoRA adapters on 30k samples without overfitting.

Why AdamW & Flash Attention 2?

AdamW over Adam

- Adam couples weight decay with gradient scaling – can over-regularize.
- AdamW **decouples** weight decay from adaptive gradients $\rightarrow$ better generalization.
- Standard for LLaMA fine-tuning; prevents over-shrinking of LoRA adapters.

Flash Attention 2 over regular / Flash Attention 1

- Standard attention: $O(N^2)$ memory – prohibitive for 577 vision tokens + text.
- Flash Attention 1 tiled computation, but under-utilised GPU parallelism.
- Flash Attention 2:
  - **Fewer non-matmul operations** $\rightarrow$ higher throughput.
  - **Better batch/head parallelism** $\rightarrow$ faster training.
  - **Lower VRAM usage (5-10$\times$ reduction)** $\rightarrow$ feasible on single GPU.

Together: Stable convergence + memory-efficient training at scale.

Results

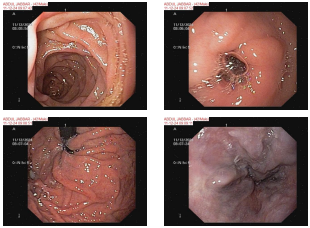
Generated Report

Medical Endoscopy Report

Patient Information

Patient Name:	ABDUL JABBAR
Patient ID:	ABDUL JABBAR_84105_RP241211_091205
Report Date:	2026-04-13 18:09:26

Endoscopy Images



Findings

Indication:	CLD-UGIB
Scope Passed Upto:	D2
Premedication:	Lignocaine spray
Esophagus:	Normal
Ge Junction:	Lax at 38cm
Stomach:	
Fundus:	Edematous mucosa noted
Body:	Edematous mucosa noted
Antrum:	Erythematous mucosa noted
Pylorus:	Normal
Duodenum:	D1: Normal | D2: Normal
Rut:	Not taken
Biopsy:	Not taken
Impression:	GRADE II ESOPHAGEAL VARICES WITH RCS EVL DONE SEVERE PHG

Report generated by Medical LLaVA AI (FFT) on 2026-04-13 at 18:09:26

Final diagnostic report – Pages 1 and 2

Quantitative Comparison

figures/comparison_table.png

Future Work

Model Optimization & Quantization

GGUF Format Conversion Wong [2024], SandLogic Technologies [2024]

- Optimizes CPU and mixed CPU-GPU inference
- Significantly improved model loading times Corp. [2024], Wong [2024]

Multi-Level Quantization Strategy SandLogic Technologies [2024], Gupta [2024]

Format	Use Case	Size Reduction
Q8_0	High-end workstations	∼50%
Q5_K_M	Standard clinical computers	∼69%
Q4_K_M	Basic clinical terminals	∼75% (3.25GB) TheBloke [2024]
Q3_K_L	Mobile deployments	Ultra-compact

Deployment Planning

Two Deployment Strategies:

1. **vLLM Framework** Fregly [2024], Linh [2024]
   - High-throughput clinical environments
   - Concurrent access scenarios
2. **Ollama Framework** Fregly [2024], Gopal [2024], Linh [2024]
   - Individual workstations
   - Low-resource environments
   - Resource-constrained deployment

EHR Integration:

- RESTful API architecture (secure, authenticated endpoints)
- Real-time synchronization with patient records

Conclusion

Key Achievements:

- Built a vision-language model for automated endoscopic report generation, fine-tuned on a custom 32,545-sample Q&A dataset.
- Implemented a DPDP-compliant de-identification pipeline, making sensitive clinical data safe for AI training.
- Integrated system prompts and Chain-of-Thought reasoning to produce **concise, transparent, and clinically interpretable** reports.
- Achieved high-quality diagnostic outputs rivaling manually written reports in structure and accuracy.

Impact: Drastically reduces documentation time; provides explainable AI assistance to gastroenterologists Eaton [2024], Hurst [2024]. **Path Forward:** Model quantization Wong [2024], SandLogic Technologies [2024] and clinical deployment trials (pending IEC approval) will bring the tool into real-world endoscopic workflows.

Acknowledgments

- **Government Medical College (GMC), Kozhikode**
  - Provided the dataset of 6,500 procedures and invaluable clinical expertise.
- **Medical Staff**
  - Continuous feedback and validation throughout development.
- **Mentors & Colleagues**
  - Project guidance and support from NIT Calicut faculty.

Thank You!

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